Is Social Learning Gendered?[*]

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Abstract

This paper investigates whether social learning is "gendered"—that is, whether individuals respond differently when given information about the behavior of own-gender versus opposite-gender peers—and examines the economic implications of such behavior. To address this question, we conduct a field experiment with a microcredit lender in Ghana. Borrowers are randomly provided with accurate information about loan requests from peers who are similar to them, with treatments varying whether this peer information comes from individuals of their same gender, the opposite gender, or both. We find that borrowers adjust their loan requests in response to peer information. However, these adjustments do not differ based on whether the peers are men or women. Additionally, varying the magnitude of the peer reference amounts does not influence borrower behavior. These results suggest that, in this context, social learning involves mechanisms beyond the simple updating of beliefs about peer behavior. Our findings contribute to understanding gender differences in credit markets and inform policies aimed at reducing economic disparities through informational interventions.

Keywords: gender, social learning, credit markets.

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1 Introduction

People frequently look to others for guidance, forming beliefs about what *similar* individuals do and using these perceptions to inform their own choices. A large literature has documented that such social learning shapes a wide range of economic and social outcomes (Duflo and Saez, 2003; Allcott, 2011; Sacerdote, 2011; Banerjee et al., 2013). At the same time, recent evidence shows that individuals often misperceive behaviors and preferences of others (Bursztyn and Yang, 2022), and that gender norms and stereotypes influence how people interpret and respond to information about others (Coffman, 2014; Bordalo et al., 2016; Bursztyn et al., 2017; Bordalo et al., 2019; Bursztyn et al., 2024).

Furthermore, a long tradition in sociology and social psychology, based on social identity theory, suggests that social learning might be "gendered". It posits that individuals are more likely to trust, recall, and act on information from in-group members, such as peers of their own gender, while discounting information from out-groups (Tajfel and Turner, 1979).[1] Consider a female entrepreneur in sub-Saharan Africa deciding whether to get a loan. She might ask: "What do other entrepreneurs like me do?" If she believes that women face similar structural barriers, such as cultural norms limiting credit access or business opportunities, information about female peers' loan requests may appear especially relevant. Yet, "like me" need not be defined by gender alone. Entrepreneurs may also consider business sector, location, experience, family circumstances, and many other attributes. In our setting in Ghana, women ask for substantially lower amounts of credit, about 35%, even conditional on observable characteristics and despite the fact that their default rates are lower. Do women ask for less because they do not want to mimic the behavior of men, or because they do not realize that men in similar economic situations ask for and receive more? Would providing objective information reduce or increase these gaps?

In this paper, we study these questions with a field experiment embedded in the call center operations of one of Ghana's largest microcredit lenders. Our approach follows three steps. First, we use extensive historical loan data to predict the typical loan amounts requested by individuals with similar characteristics. Second, we randomly provide loan applicants with objective information about these predicted amounts, varying whether the

[1] Moreover, cultural norms and gender roles can shape perceptions of competence and credibility, sometimes leading women and men to place greater weight on peers of their own gender (Ridgeway, 2011). Experimental studies further show that individuals frequently rely on signals that align with their identity or reinforce existing stereotypes (Eagly and Karau, 1991; Ibarra, 1993).

reference group is composed of peers of the applicant's own gender, the opposite gender, or both. Third, we measure whether applicants revise their loan requests after receiving the information. By randomly assigning the type of peer information each applicant receives, we test how learning about "people like me" influences choices, and whether gender itself shapes this process.

To understand how perceptions about others' behavior influence individual choices, and what this means for gender gaps, is challenging. People define similarity along many dimensions, not just gender, and groups may behave similarly because they share common experiences or environments rather than because they learn from each other. The key question is how much gender itself matters. If social learning is strongly gendered, individuals would mainly imitate same-gender peers. Alternatively, if individuals primarily care about similarity beyond gender, the gender of peers may matter little. Correcting misperceptions with objective information could then change behavior, but the direction and size of this effect would depend on what people initially believe and on whether they value information about the relevant reference group.

Before examining how information influences behavior, we first document entrepreneurs' initial perceptions about others' loan requests. We ask applicants directly how much they think peers with similar characteristics typically request, comparing their guesses to our model predictions, which are based on extensive historical data. We find substantial misperceptions: many people underestimate their peers' loan requests by 30–40%, while many others overestimate by similar amount or more. Overall, more people overestimate rather than underestimate. Surprisingly, individuals are no better at predicting the behavior of same-gender peers compared to opposite-gender peers. This may be because loan requests are typically private decisions, and entrepreneurs rarely share this information openly, even within gender groups. These widespread misperceptions suggest that there is significant potential for interventions that correct beliefs through objective information, allowing us to explore how such corrections influence actual borrowing decisions.

Our analysis then proceeds in two stages to establish causal effects. First, we consider a straightforward reduced-form question: Does providing entrepreneurs with accurate information about the loan requests of similar peers affect their own borrowing decisions? The answer is yes. At the end of loan application calls, we randomly assign applicants to receive objective information about the typical loan amounts requested by peers with similar characteristics. Applicants can then revise their original loan requests if they wish. On average, receiving this peer information significantly influences borrowing behavior: applicants who receive this information are 6–7 percentage points more likely to adjust their requested loan amounts, mostly upwards. Interestingly, the treatment effects do not differ based on whether applicants learn about the behavior of peers of their own gender, peers of the opposite gender, or both. All three information conditions yield essentially identical results. Thus, entrepreneurs clearly respond to information about the typical behavior of similar individuals, but we find no evidence that gender itself matters significantly. Additionally, we do not detect meaningful heterogeneous treatment effects by applicant gender, although our sample size limits our statistical power to identify such differences clearly. Still, interpreting this as evidence of entirely gender-neutral learning would be premature. Fully disentangling how gender affects social learning requires experimental variation designed explicitly to isolate gender from other characteristics.

Our experimental approach allows us to isolate the underlying mechanisms of gendered social learning. Specifically, we leverage a strategy to experimentally vary the objective peer information provided, while holding constant the characteristics, such as business sector, location, or experience, that define similarity. We train two distinct prediction models, one linear (OLS) and one nonlinear (XGBoost), using historical loan data. Both models have similar out-of-sample performance, but crucially, they yield slightly different predictions for any given applicant, with one model typically predicting a higher loan amount and the other a lower amount. We then randomly assign each applicant to receive this "high" or "low" prediction. This generates exogenous variation in the objective reference points that applicants receive, while keeping all other applicant characteristics constant. Using this setup, we employ an instrumental-variable approach, where the outcome in the first stage is the predicted loan amount in the peer group. This allows us to estimate the marginal behavioral response to, on average, a 10%-15% difference in the objective reference point.

Using this method, we find no evidence of gendered social learning. In fact, applicants do not systematically adjust their loan requests based on whether they receive a high or low reference amount, implying that these objective reference points themselves do not drive any behavioral response. We also check whether the marginal response depends on whether people initially underestimate or overestimate peer behavior, but find no significant heterogeneity when pooling genders. Splitting the sample by gender also does not change the conclusion, although women dominate our sample, which limits power for male-only estimates. Taken together, these findings suggest that social learning, at least in this context, does not operate through changes in expected values of beliefs about peer behavior. Instead, the behavioral response to peer information must operate through other mechanisms. We can only speculate what those may be, and in the discussion section we examine potential mechanisms based on the literature, such as cognitive uncertainty, myopia, reference points, and higher-order beliefs.[2]

Our results have policy implications for gender equality. They also speak to how a better information environment overall will influence market outcomes. First, when providing objective information to both men and women, on both genders, the gender "ask gap" effectively remains constant. This is due to the fact that such information induces similar behavioral change among both genders. By contrast, only providing information to women, regardless of reference group, effectively reduces gender gaps as it induces them to ask for more. This begs the question what is socially optimal, which is beyond the goal of this paper. We do note, however, that the answer would partly lie in the efficiency of the functioning of credit markets, such as how it may influence adverse selection or moral hazard. To that end, we estimate effects on downstream default rates in the administrative data of the company, but find no negative effects from any treatment, for any group.

We contribute to several strands of literature. First, we connect to the economics literature on how information and beliefs shape economic decisions, including studies of social learning in various contexts (Manski, 1993; Sacerdote, 2011). Recent experimental evidence shows that individuals often heavily discount information discovered by others—both within households and outside of them—relative to information they generate themselves (Conlon et al., 2021, 2022). This pattern suggests that even when conditions for efficient social learning appear to be met, individuals fail to incorporate others' signals fully, reinforcing the complexity of information integration and belief formation. Second, we speak to a broad interdisciplinary literature from sociology, social psychology, and organizational behavior that emphasizes how gender identity and cultural norms influence individuals' interpretation and use of information (Tajfel and Turner, 1979; Eagly and Karau, 1991; Ibarra, 1993; Ridgeway, 2011). Together, these literatures suggest that social learning should be systematically gendered, yet our findings show that once individuals receive precise and reliable data, these anticipated gendered differentials may not emerge.

[2]For example, Enke and Graeber (2023) discuss how cognitive uncertainty affects decision-making and can lead individuals to respond to new information in ways not predicted by standard Bayesian models. In our context, learning that others make similar requests might reassure entrepreneurs rather than altering their expected values, prompting a change in behavior without shifting mean beliefs.

Third, our results engage with research on how gender differences in "asking" and negotiation contribute to persistent gaps in economic outcomes (Babcock et al., 2003; Leibbrandt and List, 2015; Reuben et al., 2017; Biasi and Sarsons, 2022). Closely related to our work, Roussille (2024) demonstrates that lower wage demands by women can explain residual gender pay disparities. While Roussille (2024) relies on a uniform (gender-neutral) reference point and thus cannot directly identify gender-specific learning parameters, our setting provides separate predicted outcomes by gender, enabling the identification of key parameters and testing whether learning differs by the gender composition of the reference group. Moreover, our analysis focuses on loan requests rather than wage demands. Unlike wages, borrowing involves future liabilities and uncertainty, raising the possibility of different dynamics in how individuals incorporate peer information. It remains unknown if similar patterns would arise in the case of wage demands, but our methodological approach could be deployed in such settings, and beyond, and provide answers to how gender gaps arise and how gender-specific information interventions can influence those gaps, as compared to gender-neutral information.

Finally, our conclusion that changes in predicted averages do not alter behavior complements recent evidence in the behavioral economics and finance literature showing that standard Bayesian models may not capture how individuals integrate signals (Gennaioli and Shleifer, 2018; Bursztyn and Yang, 2022; Enke and Graeber, 2023). Methodologically, our field experiment—introducing exogenous variation in gender-specific predicted outcomes—broadens the empirical toolkit for identifying learning parameters. To our knowledge, no prior field work has isolated parameters like ρ_g and ρ_{-g} (gendered-based learning parameters, as we discuss below) in a real-world setting. By doing so, we offer a new lens for understanding how beliefs, social identity, and objective data interact, informing both theory and policy in contexts where decision-makers grapple with complex information drawn from diverse peer groups.

2 Experimental Design

2.1 Conceptual Framework

Our empirical strategy leverages an experimental design that directly elicits individuals' beliefs about the distribution of outcomes y—in our case, the loan amounts that "similar" entrepreneurs request—and then exogenously alters those beliefs through randomized information treatments. Let $g \in \{M, F\}$ denote an individual's gender and $-g$ the other gender. For each individual i of gender g, $y_{i,g}$ is the chosen loan amount, and $\hat{y}_{-i,g}$ and $\hat{y}_{-i,-g}$ represent the individual's beliefs about the average requested amounts among same-gender and opposite-gender peers who share their characteristics X_i. In the simplest case, we then have:

$$y_{i,g} = \alpha + \rho_g \hat{y}_{-i,g} + \rho_{-g} \hat{y}_{-i,-g} + \delta X_i + \varepsilon_i. \tag{1}$$

Here, X is a vector of characteristics (e.g., sector, location, age) that not only influence behavior directly, but also define similarity beyond gender, shaping what "people like me" means. To that end, $\hat{y}$ captures conditional expectations given X.[3] The expression captures beliefs about those who share the same characteristics, and ρ_g and ρ_{-g} capture how gender-based learning interacts with these other dimensions of similarity. For example, if social learning is gendered, then $0 < \rho_g > \rho_{-g}$. If beliefs about "people like me" matter, but not gender *per se*, then $\rho_g = \rho_{-g} > 0$. Importantly, due to misperceptions, what people believe about $\hat{y}_{-i,g}$ or $\hat{y}_{-i,-g}$ may be distorted for various reasons. Providing objective information could then have a causal effect on demand for credit, but the direction of the effect would depend on priors and parameters. For example, if women know what other women do, but do not realize that men typically ask for more in similar economic situations, providing such information to women could reduce gender gaps, but only if $\rho_{-g} > 0$. However, if $\rho_{-g} = 0$, they would ignore such information and gender gaps in outcomes would remain.[4] There are many possible scenarios. To identify parameters, one needs not only exogenous shifts in beliefs for each reference group independently, but also a consistent way to define the notion of similarity.[5]

[3] That is, $\hat{y}_{-i,g} = E[y_{j,g} \mid X_j = X_i]$ and $\hat{y}_{-i,-g} = E[y_{j,-g} \mid X_j = X_i]$ for $j \neq i$. A subtle but important distinction here is the expectations operator. Above, it refers to perceptions, i.e. subjective beliefs, which may differ from objective means. But if subjects had full access to ground-truth data, and faced no cognitive constraints, subjective and objective expectations would be the same. This paper studies reactions to *objective* data. Also note that this model could accommodate asymmetric cases; for instance, women may have different parameters than men.

[4] In the framework, gaps can arise independent of beliefs because of distributional differences in X, or because the constant is specific to a gender, α_g. Moreover, the effect of X_i can in principle differ by gender, which is what we in practice allow for empirically by estimating separate regression models for men and women.

[5] This echoes the well-known reflection problem and the identification challenges highlighted by Manski (1993), which arise when the average behavior of a group is directly *observable*. These challenges are driven by endogenous effects (how individual behavior responds to peer behavior), contextual effects (influences from group characteristics), and correlated effects (shared shocks or unobserved factors). Without experimental variation, even with perfect observation of peer behavior, it is difficult to determine whether changes in

In our field experiment, subjects are loan applicants contacting the partner microcredit institution's call center to initiate their loan applications. Before any information is revealed to them, we ask each subject to report their beliefs about what "similar" entrepreneurs—those with the same vector of characteristics X_i and a specific gender—request on average. Specifically, we elicit their prior beliefs:

$$\hat{y}^{(prior)}_{-i,g} \quad \text{and} \quad \hat{y}^{(prior)}_{-i,-g},$$

Next, we provide the entrepreneurs with exogenous information in the form of model-predicted loan amounts. Using a rich dataset of historical loan applications and outcomes from the lending institution, we train predictive models separately by gender to produce estimates[6]:

$$\tilde{y}_g(X_i) \quad \text{and} \quad \tilde{y}_{-g}(X_i).$$

These model-based predictions approximate the conditional expectations $E[y_{j,g} \mid X_j = X_i]$ and $E[y_{j,-g} \mid X_j = X_i]$, which, if accurate, represent "objective" benchmarks for what peers with similar characteristics actually request. We treat these predictions as signals that may update the subject's beliefs.

We implement three randomized treatment arms to identify the causal effect of changing beliefs about own-gender and opposite-gender (similar) peers. Each subject in a call with the microcredit institution to request a loan is randomly assigned to one of four groups:

1. **Control**: The subjects report $\hat{y}^{(prior)}_{-i,g}$ and $\hat{y}^{(prior)}_{-i,-g}$ and receive no additional information. Subjects are then asked whether they would like to change their initial request.

2. **Own-gender Information**: After eliciting $\hat{y}^{(prior)}_{-i,g}$, we provide subjects with information $\tilde{y}_g(X_i)$, but no information about the other gender. Subjects are then asked whether they would like to change their initial request.

3. **Opposite-gender Information**: After eliciting $\hat{y}^{(prior)}_{-i,-g}$, we provide subjects with information $\tilde{y}_{-g}(X_i)$, but no information about their own gender. Subjects are then asked whether they would like to change their initial request.

individual behavior are caused by genuine learning or by shared environments and common shocks. The presence of more complex or asymmetric learning patterns further exacerbates these difficulties.

[6] We provide details of how we generate these predictions in Section 2.3 below

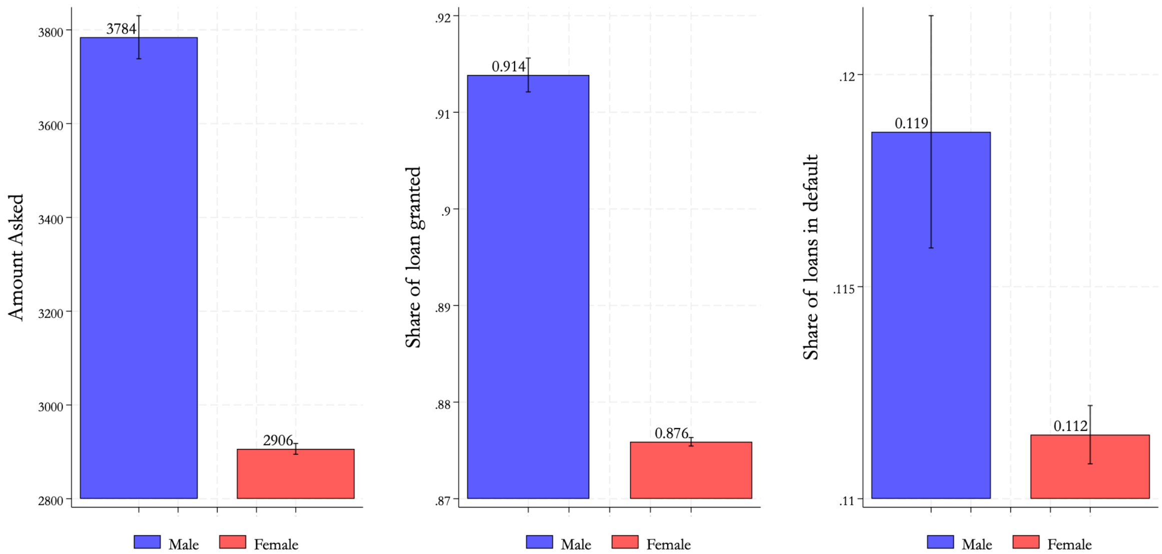
4. **Both-gender Information**: After eliciting both $\hat{y}^{(prior)}_{-i,g}$ and $\hat{y}^{(prior)}_{-i,-g}$, we provide subjects with both pieces of information: $\tilde{y}_g(X_i)$ and $\tilde{y}_{-g}(X_i)$. Subjects are then asked whether they would like to change their initial request.

By comparing how subjects update their initial loan requests relative to the control group, we can infer whether the introduction of gender-specific benchmarks influences their choices. Random variation in which signals they receive (own-gender, opposite-gender, or both) allows us to test whether ρ_g and ρ_{-g} differ, thereby revealing whether social learning is gendered. There are two reasons why we include the both-gender information treatment. First, it helps us understand whether individuals combine information about own-gender and opposite-gender peers additively or through some other mechanism. Second, providing information exclusively about one gender could inadvertently influence beliefs about the other gender, a spillover we cannot directly measure because explicitly eliciting such cross-gender updating felt unnatural in our experimental context. Thus, providing objective information about peers of both genders helps us "fix" beliefs about peers of both genders, and enables us to test whether $\tilde{y}_{g,bothgenders}(X_i)$ and $\tilde{y}_{-g,bothgenders}(X_i)$ differ from $\tilde{y}_{g,singlegender}(X_i)$ and $\tilde{y}_{-g,singlegender}(X_i)$.

Additionally, we introduce extra variation in the level of the provided signal by running two predictive models (a linear OLS and a non-linear XGBoost model) and randomly selecting which prediction (either the "high" or the "low") to provide. Both use the same X_i for prediction and have similar out-of-sample performance, as it is reported in Appendix Table B.2, but can yield slightly different estimates for each individual. This ensures an experimental variation not only in whether subjects receive any information/signal, but also in the signal's magnitude. This helps identify how sensitive subjects are to shifts in their perceived norms and, ultimately, whether their behavior aligns with standard Bayesian updating or is driven by other mechanisms.

In summary, the experimental design directly measures priors, provides exogenous signals regarding average requested amounts conditional on X_i and gender, and then observes how subjects adjust their own loan requests. By doing so, we circumvent standard identification challenges, ultimately enabling us to test whether $\rho_g = \rho_{-g}$ and, more broadly, whether social learning is gendered in our context.

Figure 1: Three Facts from the Initial Loan Book Analysis



a) Ask Gap b) Granted Gap c) Default Gap

Notes: Panels A to C present regression-adjusted predictions rather than simple averages, by gender of the loan applicant. Panel (a) shows the predicted amount requested after controlling for loan cycle, marital status, age, date, sector, and zone. Panel (b) presents the predicted share of the requested amount granted, with additional controls for the amount requested. Panel (c) depicts the predicted share of loan applicants who defaulted for more than four weeks, controlling further for both the amount requested and the amount granted. Data are based on 350,000 loan applications received from our partner institution. Gray bars indicate 95% confidence intervals.

2.2 Setting

Our study is conducted in partnership with one of the largest microcredit institutions in Ghana, a setting well-suited to studying gender differences in access to credit. Ghana is known for its vibrant entrepreneurial culture, especially among women (Hill, 1984). Approximately 82% of Ghanaian women in the workforce are self-employed (World Bank, 2022), a substantially higher share than in comparable sub-Saharan African (SSA) and lower-middle-income countries (LMICs). Consistent with this, Ghanaian women use business credit at higher rates (12.5%) than their counterparts in other LMICs (7.2%) and SSA countries (10.8%)(World Bank, 2023). Microcredit institutions are crucial sources of funding for female entrepreneurs running small businesses, predominantly in retail and food sectors.

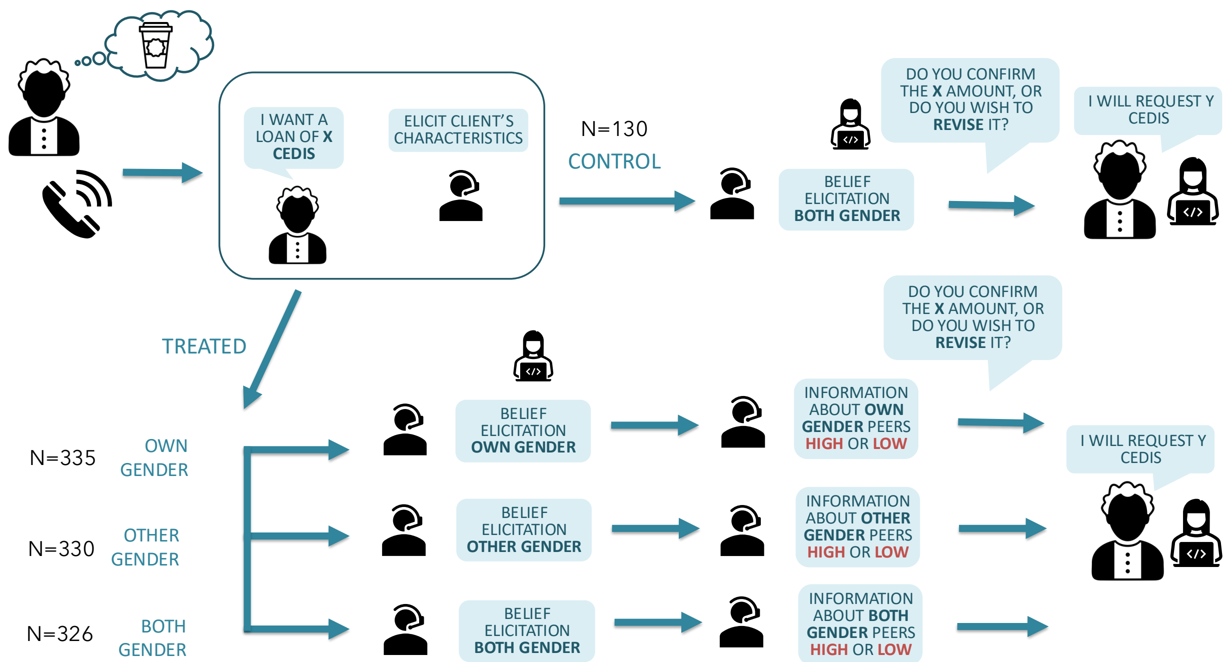
Our partner microcredit institution lends a majority (81%) of its loans to women. The median loan size is GH₵2000 (about US$250 at market exchange rates), and 95% of loans are smaller than GH₵4000. The loans are most commonly disbursed to small traders and food vendors (see Appendix Figures A.5 and A.6). The institution disburses approximately 1000 new loans per week. Based on extensive historical records we obtained from it, comprising approximately 365,000 loan applications and information on defaults between May 2020 and June 2022, we document three important facts, which we summarize in Figure 1. We find that over the 26-month horizon analyzed: (1) male borrowers request loans that are on average 30% larger than those requested by female borrowers; (2) male borrowers are granted a share of their initial request that is 3.8 percentage points higher than that of women; (3) the share of men who are more than four weeks in default is 0.7 percentage points higher than for women. Together, these facts reveal a striking pattern: the are substantial "ask" gaps in loan requests between male and female borrowers, yet women outperform men on repayment. This misalignment between creditworthiness and credit access suggests that the gap is unlikely to be driven by fundamentals, and points instead to demand-side underreach among women as one possible explanation. Our intervention targets precisely the demand margin by providing women with information about peers' average loan requests during the application process.

The loan application process at the microcredit institution involves five steps. First, potential borrowers contact the call center to initiate a loan application. Second, call center agents register the application, identify the borrower's business location, and forward the information to the corresponding team of field officers for that area. Third, these field officers conduct an in-person visit to the applicant's business to assess creditworthiness and recommend an amount to be granted. Fourth, credit analysts at headquarters review the recommendation and finalize the approved loan amount. Finally, the loan is disbursed via mobile money, usually within one week of the approval date. The working capital loans analyzed in this paper mature after 13 weeks and require repayment in equal weekly installments, beginning one week after disbursement, with an interest rate of 12% per month. We summarize the loan application process in Appendix Figure A.1.

Our experimental intervention, detailed in Section 2.3, takes place during steps 1 and 2 of this process, specifically during the initial loan application stage at the call center. In the final section of the results, we further analyze the loan amounts ultimately approved by credit analysts (step 4), as well as borrowers' repayment performance throughout the 13-week repayment period.

2.3 Design

Figure 2: Experimental Design



Notes: The figure shows the sequence of the loan application and embedded experiment. All applicants first stated an initial loan request and had their prior beliefs about peers' requests elicited. Treated applicants then received predicted average loan amounts for peers of their own gender, the opposite gender, or both (with otherwise identical characteristics), while the control group received no information. Predictions were generated by XGBoost and OLS models estimated separately by gender on historical data; the highest or the lowest of the two was randomly selected per applicant, inducing exogenous variation in the magnitude of the information provided. All applicants were then given the opportunity to revise their initial request. Assignment probabilities were 0.3 per treatment arm and 0.1 for the control group.

Our experiment was conducted at the call center of the microcredit institution in Accra, Ghana, which is the first point of contact for potential borrowers when applying for a loan. The details of the experimental design are illustrated in Figure 2. Upon receiving a loan application call, the call center agents first recorded the initial amount requested by each client. The agent then collected detailed client information, including name, gender, age, marital status, business type, and business and residence locations. After registering these details, the agent elicited the client's priors about peer behavior by asking how much she believed a typical male or female borrower with identical characteristics to her would typically request at the same company.[7]

After prior elicitation, the information intervention followed: Clients were randomly assigned to either the control group or one of three treatment arms. Treatment clients received objective information about typical loan amounts requested by own-gender, opposite-gender, or both-gender borrowers with identical characteristics, while control clients received no such information.[8] In the last step, clients in both treatment and control groups were asked if they wished to revise their initial loan requests. This final loan request was then recorded by the agent. We provide the full call script that agents followed in Appendix D.1.

The information provided to clients during the experiment was based on predicted average loan amounts requested by peers with their same characteristics. Predictions were generated using two distinct models: an XGBoost (Extreme Gradient Boosting) machine learning model and a standard Ordinary Least Squares (OLS) model.[9] Both models were trained on historical loan application data from the microcredit institution, covering 280,866 loan requests made between January 2021 and June 2022, after excluding outliers below the 5th and above the 95th percentile of requested amounts. Among these loan applicants, 79.7% were women. We present a summary of the training dataset's characteristics in Appendix Table B.6.[10] To ensure predictions were informative to participants, we restricted the experiment to women requesting amounts below GH₵9,500 (the weighted average upper limit of the training data) and in their ninth loan cycle or earlier. Both the OLS and XGBoost models were estimated separately for men and women, resulting in four distinct predictions for each client profile.[11] The relevant predictions (either own-gender, opposite-gender, or both-gender) of only one of the two models was randomly chosen to be communicated to the client. This strategy allowed us to introduce exogenous variation in the magnitude of the information treatment provided to each participant. Specifically, for each participant,

[7] When eliciting priors, clients in the treatment groups were asked only about the gender(s) for which they subsequently received information. Clients in the control group were asked about both genders, although they received no further information.

[8] The probability of assignment to the control group was 0.1, while assignment to the treatment group had a probability of 0.9 (with equal probabilities of 0.3 for each treatment arm). This resulted in a relatively smaller control group compared to the treatment groups.

[9] Predictions were adjusted for inflation, which was computed using the monthly CPI reported by the Central Bank of Ghana (base year: 2018).

[10] We classified occupations into sectors using OpenAI's GPT-3.5 Turbo model, as detailed in Appendix C.

[11] The two models exhibited comparable predictive performance, as reported in Appendix Table B.2.

we ranked the predictions from the two models (a high and a low one, but based on the same set of observable characteristics) and only one was communicated to the client.[12] This design will enable us to employ an instrumental variable regression approach, as discussed below, using the randomized choice between high and low predictions as an instrument to investigate how the magnitude of the information provided influenced participants' behavior.

3 Data, Outcomes and Empirical Strategy

3.1 Data

Our experiment involved 1,113 loan applicants, and ran over three consecutive weeks between November and December 2023. As shown in Appendix Table B.1, 76.5% of participants are women, reflecting the company's predominantly female client base. Around 70% of respondents are married, 24% are single, and the remainder are separated, divorced, or widowed. The average age of participants is 38.9 years. In terms of loan experience, 39.2% are applying for their first loan with the institution, 23.4% are on their second loan, and 37.4% have taken three or more loans previously. The most common sectors of activity are food selling (36.1%) and retail and non-food trade (29.9%).

The initial loan amount requested averages GH₵3,347 across the full sample, with broadly similar averages across treatment arms. All baseline characteristics are well-balanced across experimental arms, and joint F-tests fail to reject the null of no systematic differences between the control group and each treatment arm (Appendix Table B.1).

For the predictive component of our design, we use approximately 280,866 historical loan applications from January 2021 through June 2022, covering the same institution. This training dataset is described in detail in Appendix C. We train separate prediction models for men and women on this dataset, allowing the predicted reference points we communicate to applicants to reflect gender-specific conditional means.

[12] The average difference in predicted amounts was between 10%–15%; see Appendix Figure A.2 for the full distribution.

3.2 Outcome Variables

We study the impact of the information treatment on several outcomes that capture whether and how applicants revise their loan requests, as well as downstream loan performance.

Any change. A dummy variable equal to 1 if the participant revises their final requested amount relative to the initial amount, regardless of direction.

Adjusted up. A dummy variable equal to 1 if the participant increases their final requested amount.

Adjusted down. A dummy variable equal to 1 if the participant decreases their final requested amount.

Log(F/I). The log difference between the final and initial amounts requested, log(Final/Initial). This captures the magnitude and direction of any revision.

Log(G/I). The log difference between the amount granted by the institution and the initial amount requested. This captures whether the information-induced revisions translate into changes in the credit extended.

Non-performing loan (NPL). A dummy variable equal to 1 if the participant has an overdue balance at a given cutoff date after our intervention—we evaluate this at two dates, February 26th and March 30th, 2024—capturing whether treatment-induced upward revisions in loan requests are associated with adverse downstream loan performance.

3.3 Empirical Strategy

Reduced-form analysis. To estimate the effect of providing information about similar peers, we estimate the following equation:

$$y_{i,g} = \alpha + \beta_1 \, \mathrm{Info}_i + X_i' \, \lambda + \gamma_c + \theta_g + \epsilon_i \tag{2}$$

where y_i is one of the outcomes described above, Info_i is a binary indicator equal to 1 if applicant i was assigned to any information treatment, X_i' is a vector of individual controls, and γ_c and θ_g denote call center agent and gender fixed effects, respectively. In our baseline specification, we always include call center agent and gender fixed effects. In specifications with full controls, we additionally absorb week, marital status, loan cycle, region, and sector fixed effects. For the binary outcomes (any change, adjusted up, adjusted down), we also include quintile-of-initial-amount-requested fixed effects. All regressions are estimated with robust standard errors.

To study whether treatment effects differ by the gender composition of the reference group, we replace Info_i with indicators for own-gender information (Own gender), opposite-gender information (Other gender), and both-gender information (Both), estimating:

$$y_{i,g} = \alpha + \beta_1 \mathbf{1}[\text{Own gender}]_i + \beta_2 \mathbf{1}[\text{Other gender}]_i + \beta_3 \mathbf{1}[\text{Both}]_i + X_i' \lambda + \gamma_c + \theta_g + \epsilon_i \quad (3)$$

The null hypothesis of gender-neutral social learning corresponds to $\beta_1 = \beta_2$: applicants respond equivalently to information about own-gender and opposite-gender peers.

Instrumental variable analysis. The reduced-form estimates above capture the effect of receiving information about similar peers. To identify whether the *level* of the reference amount matters—i.e., whether applicants respond to a higher or lower predicted peer amount—we exploit the random assignment of the prediction model as an instrument.

As described in Section 2, we train two models—OLS and XGBoost—separately by gender. For any given applicant with characteristics X_i, the two models generate different predictions, with one being a higher amount and the other a lower. We randomly assign each applicant to receive the high or low prediction (dummy High_i). The log of the communicated amount, $x_i = \log(\text{AmountSaid}_i)$, therefore varies exogenously across applicants with identical characteristics X_i. We use High_i as an instrument for x_i in a 2SLS specification:

$$x_{i,g} = \delta + \pi \,\text{High}_i + X_i' \mu + \gamma_c + \theta_g + u_i \quad (4)$$

$$y_{i,g} = \alpha + \rho \, x_i + X_i' \lambda + \gamma_c + \theta_g + \epsilon_i \quad (5)$$

where ρ captures the marginal behavioral response to a 1% increase in the communicated peer amount, holding the applicant's characteristics fixed. This specification is estimated on the single-treatment sample (own-gender or opposite-gender only), since the both-gender arm involves two communicated amounts and four possible combinations of high and low. We also interact x_i with an indicator for opposite-gender treatment ($x_i \times \mathbf{1}[\text{Other}]_i$), instrumented by $\text{High}_i \times \mathbf{1}[\text{Other}]_i$, to test whether the marginal response to the communicated amount differs by reference group gender. All 2SLS specifications are estimated with robust standard errors and the Kleibergen-Paap F-statistic is reported to assess instrument strength.

4 Results

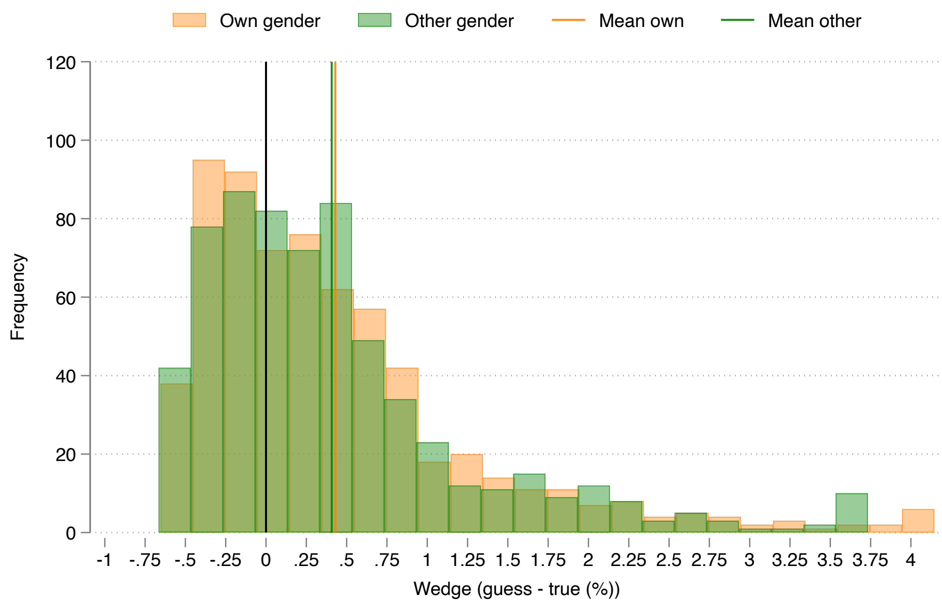
4.1 Beliefs

Before turning to treatment effects, we first document the baseline accuracy of entrepreneurs' perceptions of peer loan requests. Prior to receiving any information, we ask applicants to estimate how much borrowers with similar characteristics—same line of work, age, region, marital status, and loan cycle—typically request at the microcredit institution. We then compare these stated guesses to the model-predicted conditional means, computing the wedge between the two as $(guess - true)/true$, where *true* is the amount the OLS or XGBoost model predicts for a borrower with those characteristics.

Figure 3 plots the distribution of these wedges for own-gender and opposite-gender beliefs, pooling all respondents who were asked about peers. The distributions reveal substantial misperceptions in both directions. Many applicants underestimate the typical request of similar peers by 30–40%, while a similar share overestimate by comparable or larger amounts; on balance, overestimation is substantially more common: roughly 64% of respondents overestimate opposite-gender peer requests and 63% overestimate own-gender peer requests. These misperceptions are not small: the average wedge is meaningfully different from zero in both directions across subgroups. Strikingly, applicants are no better calibrated about the behavior of own-gender peers than about opposite-gender peers. The distributions for own-gender and other-gender beliefs closely overlap, and the mean wedges are not statistically distinguishable. This is perhaps unsurprising given that loan requests are private—applicants rarely observe peers' borrowing behavior directly, and there is no systematic information channel organized by gender. Appendix Figures A.3 and A.4 show the analogous patterns separately for female and male respondents and confirm that this finding holds within each gender group.

Together, these patterns indicate that misperceptions are widespread, symmetric across genders as reference groups, and leave ample scope for the information treatment to shift beliefs and behavior.

Figure 3: Wedges in Perceptions of Similar Peer's Amount Requested



Notes: This figure plots the distribution of misperceptions about the loan amounts requested by similar peers. The wedge is defined as (guess − true)/true, where *guess* is the respondent's prior belief about the average amount requested by borrowers with the same characteristics as her, and *true* is the model-predicted amount conditional on the respondent's characteristics. *Own gender* refers to beliefs about peers of the same gender as the respondent; *Other gender* refers to beliefs about peers of the opposite gender. The sample includes all respondents who received information about peers of their own gender, opposite gender, or both. Vertical lines indicate the sample mean for each group. Wedges are winsorized at the 1st and 99th percentiles.

4.2 Reduced-form Effects

Table 1 presents the reduced-form estimates of equation (2). It shows that applicants who receive any peer information are between 6.1 and 6.6 percentage points more likely to revise their request, relative to a control mean of 4.6%. Most revisions are upward: the probability of increasing the own request rises by 5.0 to 5.4 percentage points. Downward revisions increase by 1.2 percentage points. The effect on *Log(F/I)* is also positive, indicating that the treatment increases requested amounts on average.[13]

Table 1: Effects of Peer Information on Borrowing Behavior

	Any change		Adjusted up		Adjusted down		Log(F/I)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Info	0.066***	0.061***	0.054***	0.050**	0.012***	0.012***	0.013*	0.012*
	(0.0212)	(0.0217)	(0.0210)	(0.0214)	(0.0037)	(0.0040)	(0.0068)	(0.0069)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Control mean	0.046	0.046	0.046	0.046	0.000	0.000	0.014	0.014
Observations	1,113	1,113	1,113	1,113	1,113	1,113	1,113	1,113

Notes: This table summarizes the main results from providing any type of information. *Log(F/I)* is the log difference between the final and initial amount asked. All regressions include call center agent and gender fixed effects . Columns (1) to (6) include quintile of initial amount asked fixed effects. Additional controls include age, region, sector, week and loan cycle fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

We then ask whether the response depends on the gender of the reference group. Table 2 shows little evidence that it does. Applicants who receive information about own-gender peers are 7.3 percentage points more likely to revise their request. Applicants who receive information about opposite-gender peers are 7.4 percentage points more likely to do so. The corresponding effects on upward revisions are 6.0 percentage points in both arms. The both-gender arm is somewhat smaller, at 4.9 percentage points for any change and 4.3 percentage points for upward revisions, but the pairwise differences between the three treatment arms are not statistically significant. The table also rejects additivity for several outcomes: receiving both signals does not generate the sum of the own-gender and opposite-gender effects. Still, the main conclusion is unchanged. Applicants respond to information about similar

[13] Appendix Table B.3 replicates this analysis interacting the treatment with applicant gender. We do not find statistically significant heterogeneous effects, though the sample is predominantly female and we have limited power to detect differential responses by gender.

borrowers, but we do not find evidence that they respond differently depending on whether the reference group is composed of their own-gender or opposite-gender peers.[14]

Table 2: Effects of Peer Information on Borrowing Behavior by Reference Group Gender

	Any change		Adjusted up		Adjusted down		Log(F/I)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Own gender	0.073***	0.068***	0.060**	0.056**	0.013**	0.013**	0.017*	0.017*
	(0.0257)	(0.0258)	(0.0252)	(0.0252)	(0.0063)	(0.0061)	(0.0094)	(0.0094)
Other gender	0.074***	0.068***	0.060**	0.054**	0.015**	0.014**	0.010	0.008
	(0.0259)	(0.0262)	(0.0251)	(0.0255)	(0.0070)	(0.0067)	(0.0082)	(0.0082)
Both	0.049**	0.047*	0.043*	0.040	0.007	0.007	0.011	0.010
	(0.0249)	(0.0258)	(0.0246)	(0.0254)	(0.0045)	(0.0049)	(0.0080)	(0.0081)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Mean dep. var.	0.106	0.106	0.096	0.096	0.010	0.010	0.026	0.026
p-val $\beta_{own} = \beta_{other}$	0.970	0.981	0.978	0.931	0.859	0.863	0.443	0.346
p-val $\beta_{own} = \beta_{both}$	0.326	0.375	0.450	0.496	0.403	0.427	0.535	0.488
p-val $\beta_{other} = \beta_{both}$	0.304	0.391	0.459	0.551	0.335	0.357	0.855	0.744
p-val $\beta_{own} + \beta_{other} = \beta_{both}$	0.006	0.011	0.024	0.040	0.045	0.039	0.203	0.249
Observations	1,113	1,113	1,113	1,113	1,113	1,113	1,113	1,113

Notes: This table summarizes the main results from the experiment. The treatment is providing information about own gender peers, opposite gender peers, or both. *Log(F/I)* is the log difference between the final and initial amount asked. All regressions include call center agent and gender fixed effects. Columns (1) to (6) include quintile of initial amount asked fixed effects. Additional controls include age and region, sector, week and loan cycle fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3 shows that the reduced-form response is concentrated among repeat borrowers. Among these more experienced clients, information increases the probability of any revision by 10.2 percentage points and increases *Log(F/I)* by 0.024. For first-time borrowers, these effects are almost entirely offset: the interaction terms are −9.7 percentage points for any revision and −0.029 for *Log(F/I)*. The estimates across individual treatment arms point in the same direction, with positive effects among repeat borrowers and negative interactions for first-time borrowers. This suggests that experience with the lender matters for how applicants use the information. Repeat borrowers may have a clearer benchmark for revising their request, while first-time borrowers may be more reluctant to revise the amount they initially asked later in the call.

[14] Appendix Table B.4 further interacts the treatment arms with applicant gender. Point estimates are imprecise and we cannot reject homogeneous treatment effects across male and female applicants.

Table 3: Effects of Peer Information on Borrowing Behavior: Heterogeneity by First-time Borrowers

	Any change		Log(F/I)	
	(1)	(2)	(3)	(4)
Info	0.102***		0.024***	
	(0.0244)		(0.0075)	
Info × New	-0.097**		-0.029*	
	(0.0465)		(0.0151)	
Own gender		0.101***		0.021*
		(0.0308)		(0.0109)
Other gender		0.121***		0.025**
		(0.0314)		(0.0097)
Both		0.082***		0.026***
		(0.0307)		(0.0096)
Own gender × New		-0.076		-0.010
		(0.0564)		(0.0207)
Other gender × New		-0.133**		-0.044**
		(0.0543)		(0.0173)
Both × New		-0.080		-0.036**
		(0.0527)		(0.0171)
Control mean	0.046	0.046	0.014	0.014
Observations	1,113	1,113	1,113	1,113

Notes: This table summarizes the main results from providing any type of information about the same gender, opposite gender, or both. It allows for heterogenity depending on whether the client is a first-time borrower (*New*=1) or not. All regressions control for age and include call center agent, gender, region, sector and week fixed effects. Columns (1) and (2) include quintile of initial amount asked fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.3 2SLS Results

The reduced-form estimates show that receiving information about similar peers changes borrowers' behavior, but they do not tell us whether the *level* of the communicated amount matters. Does receiving a higher reference point, for example, hearing GH₵4,000 rather than GH₵3,500 lead applicants to request more? We answer this using the instrumental-variable strategy described in Section 3.

Table 4 presents the 2SLS results for applicants who received information about a single gender. The first stage is strong across all specifications, with first-stage F-statistics of 26.8 for the binary outcomes and 77.5 for *Log(F/I)*, well above conventional thresholds. Column (1) shows that the coefficient on *log(Amount Said)* is 0.325 and significant at the 10% level. A 10% higher communicated amount would therefore increase the probability of any revision by about 3.3 percentage points. This estimate is imprecise, and the other outcomes do not show statistically significant responses. The interaction with opposite-gender information is also small and insignificant. Thus, we do not find evidence that applicants respond more strongly to higher reference amounts when those amounts come from own-gender rather than opposite-gender peers.[15]

Overall, these estimates suggest that the reduced-form effect is not mainly driven by the precise value of the reference amount. The presence of information is what matters, perhaps by reducing uncertainty or making revision feel more acceptable, ultimately leading applicants to reconsider their initial request. The data leave room for a modest effect of the level on the probability of any revision, but there is no consistent response across outcomes.

4.4 Heterogeneity by Prior Accuracy

A natural question is whether the response to the communicated amount depends on applicants' prior beliefs. Applicants who underestimated peer loan requests might respond more to a higher reference amount, while those who overestimated might respond less or revise in the opposite direction. Table 5 tests this by interacting *log(Amount Said)* with an indicator for whether the applicant's prior exceeded the model prediction.

The results provide no support for this hypothesis. Among applicants who underestimated peer requests, the coefficient on *log(Amount Said)* is small and statistically insignif-

[15] Appendix Table B.5 presents 2SLS estimates allowing the marginal response to vary by applicant gender. The estimates are imprecise and we find no evidence of differential responses.

Table 4: 2SLS Effects of Amount Said on Borrowing Behavior

	Any change		Adjusted up		Adjusted down		Log(F/I)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
log(Amount Said)	0.325*	0.239	0.262	0.252	0.063	-0.013	0.043	0.086
	(0.1957)	(0.3013)	(0.1841)	(0.2853)	(0.0706)	(0.1069)	(0.0754)	(0.1308)
log(Amount Said) × Other gender		0.156		0.017		0.139		-0.079
		(0.4144)		(0.3891)		(0.1519)		(0.1610)
First-stage F-stat	26.8		26.8		26.8		77.5	
Kleibergen-Paap F-stat		49.1		49.1		49.1		51.2
Observations	662	662	662	662	662	662	662	662

Notes: This table presents the results of a 2SLS regression, where the amount said is instrumented by the assignment to a high amount among individuals who received information about a single gender. *Log(F/I)* is the log difference between the final and initial amount asked. All regressions include controls for age, and call center agent, gender, region, sector, week and loan cycle fixed effects. Columns (1) to (6) include quintile of initial amount asked fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

icant across outcomes. The interaction for overestimation is also statistically insignificant. When we further allow these effects to vary by the gender of the reference group, the estimates remain uninformative and the first stage becomes weak, with a Kleibergen-Paap F-statistic of 4.3. We interpret these results as showing no evidence of heterogeneity by prior accuracy, while recognizing that we have limited power for these interaction specifications.

Table 5: 2SLS Effects of Amount Said on Borrowing Behavior: Heterogeneity by Under/Overestimated

	Any change		Adjusted up		Adjusted down		Log(F/I)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
log(Amount Said)	0.030	-0.169	0.117	-0.145	-0.087	-0.024	0.131	0.091
	(0.3615)	(0.6583)	(0.3480)	(0.6543)	(0.0969)	(0.0710)	(0.1263)	(0.2581)
log(Amount Said) × Overestimated	0.423	0.527	0.180	0.508	0.243	0.019	-0.167	-0.033
	(0.4639)	(0.7840)	(0.4366)	(0.7624)	(0.1607)	(0.1980)	(0.1697)	(0.3217)
log(Amount Said) × Other gender		0.308		0.410		-0.102		0.067
		(0.7603)		(0.7456)		(0.1496)		(0.2904)
log(Amount Said) × Overestimated × Other gender		-0.113		-0.546		0.433		-0.259
		(0.9240)		(0.8846)		(0.2859)		(0.3613)
Kleibergen-Paap F-stat	28.4	4.3	28.4	4.3	28.4	4.3	29.8	4.3
Observations	657	657	657	657	657	657	657	657

Notes: This table presents the results of a 2SLS regression, where the amount said is instrumented by the assignment to a high amount among individuals who received information about a single gender. *Overestimated*=1 if the individual has priors that are strictly higher than the amount predicted by the ML algorithm. *Log(F/I)* is the log difference between the final and initial amount asked. All regressions include controls for age, and call center agent, gender, region, sector, week and loan cycle fixed effects. Columns (1) to (6) include quintile of initial amount asked fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.5 Effect on Granted Amounts and Default

A final concern is that encouraging applicants to ask for more could lead to larger loans and worse repayment outcomes, potentially harming borrowers. We examine this using the lender's administrative data.[16]

Table 6 shows no statistically significant effect of receiving information about peers' requests on the amount granted. The treatment does not change the probability that the granted amount differs from the initial request, nor does it change *Log(G/I)*. This suggests that the lender's underwriting process absorbs the changes in requested amounts and does not mechanically pass them through to granted credit. This is consistent with the high control mean for any change between the granted amount and the initial request (43.8%), which shows that loan officers frequently adjust requests as part of the normal credit assessment process, typically downwards. Our intervention does not appear to meaningfully affect that assessment.

Finally, Table 7 shows no detectable effect on repayment. In Panel A, the treatment has no statistically significant effect on the probability of being classified as a non-performing loan at either cutoff date. In Panel C, the effects on principal overdue and days overdue are small in magnitude and statistically insignificant. Panel B instruments upward revision with the information treatment, but these estimates are imprecise and the first stage is weak, so they should be interpreted cautiously. Taken together, the administrative data provide no evidence that the information treatment worsened repayment outcomes.

[16]Not all experimental participants appear in the loan book data, either because their loan was not approved or because of record-matching issues. Appendix Table B.7 shows that observable characteristics are balanced between participants found in the loan books and those not found, suggesting that attrition from the administrative data is unlikely to bias our estimates.

Table 6: Effects of Peer Information on Granting Behavior

	Any change		Adjusted up		Adjusted down		Log(G/I)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Info	-0.020	-0.017	0.007	0.003	-0.027	-0.020	0.038	0.035
	(0.0497)	(0.0465)	(0.0279)	(0.0280)	(0.0472)	(0.0444)	(0.0343)	(0.0326)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Control mean	0.438	0.438	0.076	0.076	0.362	0.362	-0.146	-0.146
Observations	929	929	929	929	929	929	929	929

Notes: This table summarizes the results of providing any type of information on downstream outcomes. Any change, adjusted up and adjusted down compare the amount granted with the initial amount asked. *Log(G/I)* is the log difference between the amount granted and the initial amount asked. All regressions include call center agent and gender fixed effects. Columns (1) to (6) include quintile of initial amount fixed effects. Additional controls include age, marital status, region, sector, week and loan cycle fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Effects of Peer Information on Loan Performance

	Panel A: Extensive margin							
	OLS				2SLS			
	NPL 26.02		NPL 30.03		NPL 26.02		NPL 30.03	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Info	0.016	0.021	-0.040	-0.042				
	(0.0425)	(0.0424)	(0.0443)	(0.0435)				
Share change up					0.282	0.395	-0.706	-0.778
					(0.7549)	(0.8016)	(0.8137)	(0.8496)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Control mean	0.741	0.741	0.724	0.724				
F-stat					6.81	5.72	6.81	5.72
Observations	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024
	Panel B: Intensive margin							
	Principal 26.02		Principal 30.03		Days 26.02		Days 30.03	
Info	-49.176	-49.533	-104.929	-100.892	-1.857	-1.909	-3.979	-3.911
	(90.0695)	(88.4266)	(107.3441)	(105.1778)	(2.2263)	(2.2199)	(3.3970)	(3.3689)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Control mean	940.235	940.235	1081.725	1081.725	25.767	25.767	44.983	44.983
Observations	1,024	1,024	1,024	1,024	1,024	1,024	1,024	1,024

Notes: This table summarizes the effects of the intervention on entrepreneurs' repayment behavior, evaluated at two cutoff dates: February 26th and March 30th, 2024. Panel A shows effects on the probability that the loan is categorized as a non-performing loan (NPL)—defined as having an overdue balance by either cutoff date. Columns (1)–(4) show OLS estimates of the effect of any information on NPL status. Columns (5)–(8) show 2SLS estimates of the effect of an upward revision in the amount asked on NPL status, instrumented with the information treatment. Panel B shows the effect of the information intervention on principal and days overdue at the two cutoff dates. All regressions include quintile of amount granted (Panel A OLS and Panel B) or amount asked (Panel A 2SLS), call center agent and gender fixed effects. Additional controls include age, marital status, region, sector, week and loan cycle fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5 Conclusion

This paper studies whether social learning is gendered. We ask whether loan applicants respond differently to objective information about own-gender and opposite-gender peers when deciding how much credit to request. To answer this question, we embed a field experiment in the loan application process of a large microcredit institution in Ghana. Applicants receive information about the typical loan amounts requested by similar peers, but the reference group used to generate this information is randomly selected from among own-gender peers, opposite-gender peers, or both. We also randomize the prediction model used to generate the information, which gives us variation in the level of the communicated reference amount while holding applicant characteristics fixed.

We find three main results. First, we document widespread misperceptions about what similar peers request. These errors go in both directions, and applicants are no better at predicting the behavior of own-gender peers than opposite-gender peers. Second, providing peer information affects behavior. Treated applicants are about 6–7 percentage points more likely to revise their loan request, mostly upward. Yet the response is very similar across own-gender, opposite-gender, and both-gender information arms. Third, we find little evidence that the level of the communicated amount drives the response. The 2SLS estimates do not show a consistent effect of higher reference amounts across outcomes, nor do they show stronger responses to own-gender than opposite-gender information. The patterns we find also do not differ by whether applicants initially over- or underestimated peer behavior.

Taken together, the results suggest that applicants respond to information about similar borrowers, but not in a way that depends strongly on the gender of those borrowers. This does not mean that gender is irrelevant for all forms of social learning. Rather, in this setting, where loan requests are private and where the information is presented as objective data, gender does not appear to be the main dimension along which applicants interpret peer behavior. The response may instead come from the presence of information itself: learning what others request may reduce uncertainty, make revision feel more acceptable, or prompt applicants to reconsider an initial amount.

Our findings speak to the broader literature on how information shapes decisions. Standard Bayesian models would predict that higher reference points generate proportionally larger responses; we find no such pattern. This is consistent with recent evidence that cognitive uncertainty and non-Bayesian processing may govern how individuals integrate new data, and suggests that the mechanisms underlying social learning in field settings may be richer and more complex than simple belief updating over expected values.

There are important caveats to our results. Our sample is predominantly female, which limits our power to study heterogeneous treatment effects by applicant gender. Some of the interacted 2SLS specifications are also demanding in terms of power, for example when splitting responses by prior accuracy and reference-group gender. We therefore cannot rule out more moderate forms of gendered learning, particularly in subgroups.

The findings are nevertheless useful for thinking about information interventions. Providing objective peer information can change borrowing behavior without worsening observed repayment outcomes in our data. But the effect does not appear to come from applicants mechanically moving toward the exact reference amount they hear. This matters for policy: information can shift behavior, but its effects may depend less on the precise statistic communicated than on how the information changes uncertainty, attention, or the perceived appropriateness of revising a choice.

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Is Social Learning Gendered?

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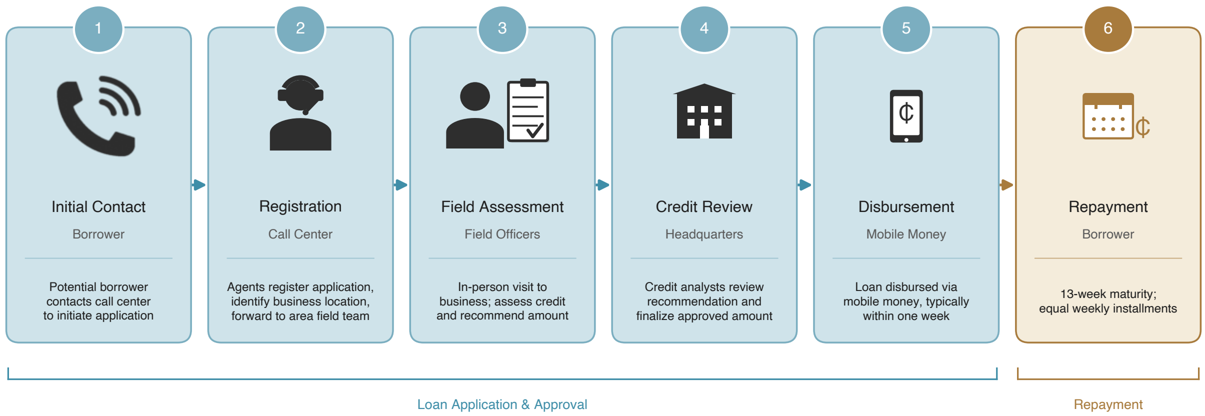
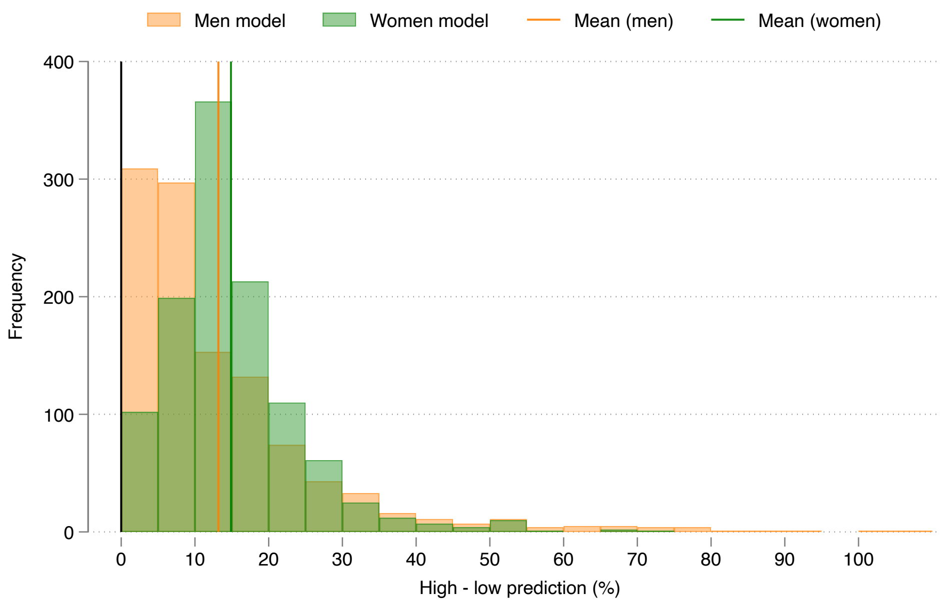
A Figures

Figure A.1: Steps of the Loan Application Process



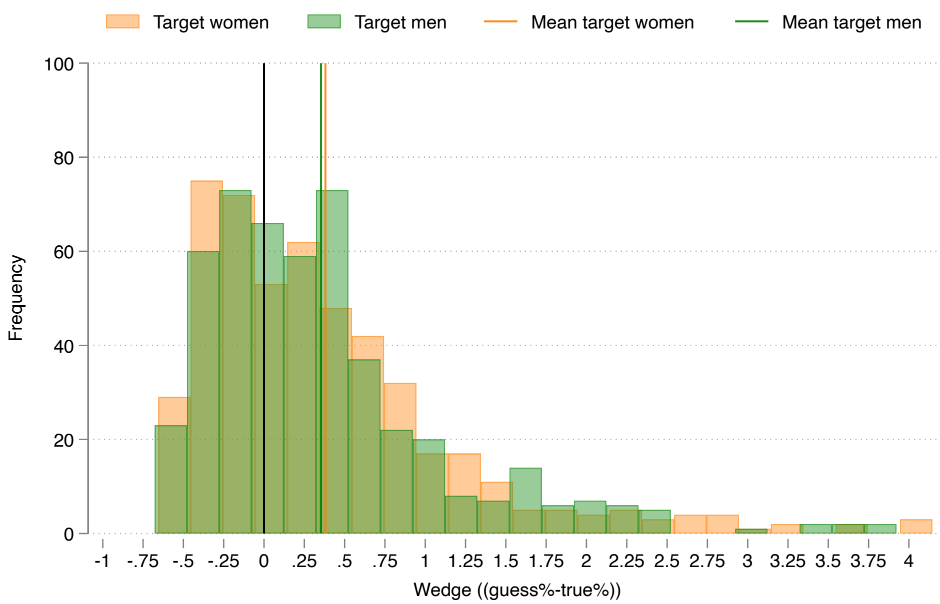
Notes: This figure outlines the loan application process at the microcredit institution. Stages 1–5 cover application and approval; stage 6 depicts the repayment schedule for the working-capital loans analyzed in the paper.

Figure A.2: Distribution of Differences Between OLS and XGBoost Predictions



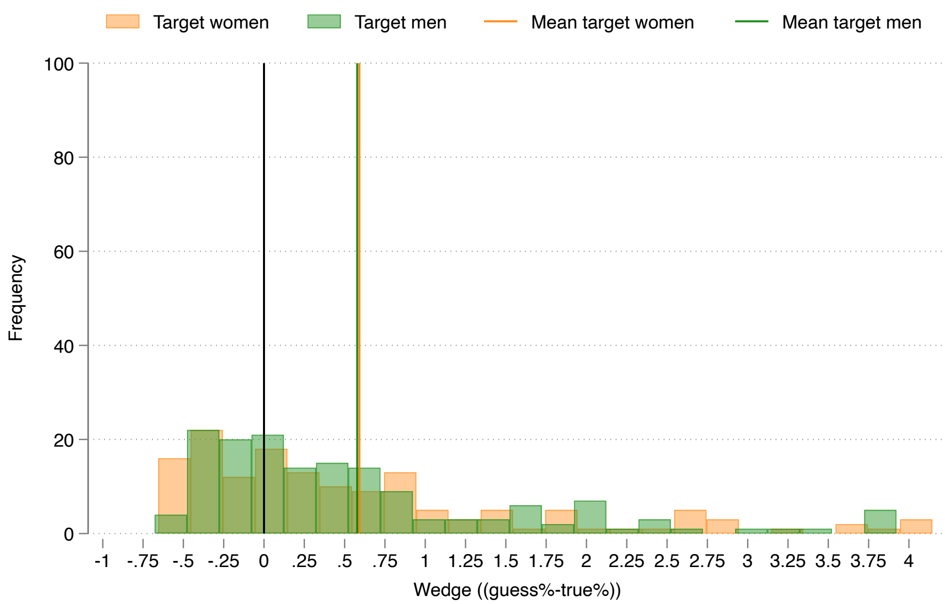
Notes: This histogram shows the distribution of absolute differences between the predicted loan amounts generated by the OLS and XGBoost models, expressed as a percentage of the lower of the two predictions. Both models were estimated separately by gender on historical loan application data from the microcredit institution. The average gap between the two predictions is 13.18% for men and 14.89% for women.

Figure A.3: Women's Wedges in Perceptions of Peer's Amount Requested



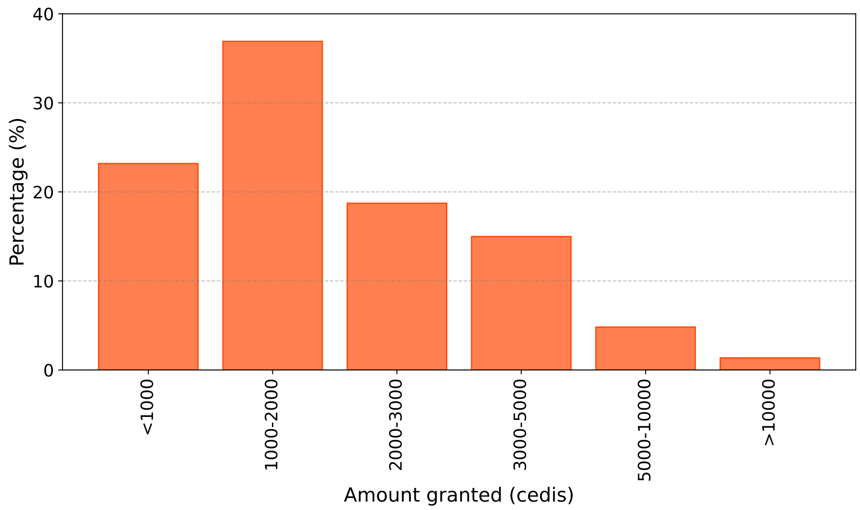
Notes: This figure plots the distribution of belief wedges among female respondents about the loan amounts requested by peers. The wedge is defined as (guess − true)/true, where *guess* is the respondent's prior belief about the average amount requested by borrowers with the same characteristics as themselves, and *true* is the model-predicted amount conditional on the respondent's characteristics. *Target women* refers to beliefs about female peers; *Target men* refers to beliefs about male peers. The sample includes all female respondents who were asked about the respective target gender. Vertical lines indicate the sample mean for each group. Wedges are winsorized at the 1st and 99th percentiles.

Figure A.4: Men's Wedges in Perceptions of Peer's Amount Requested



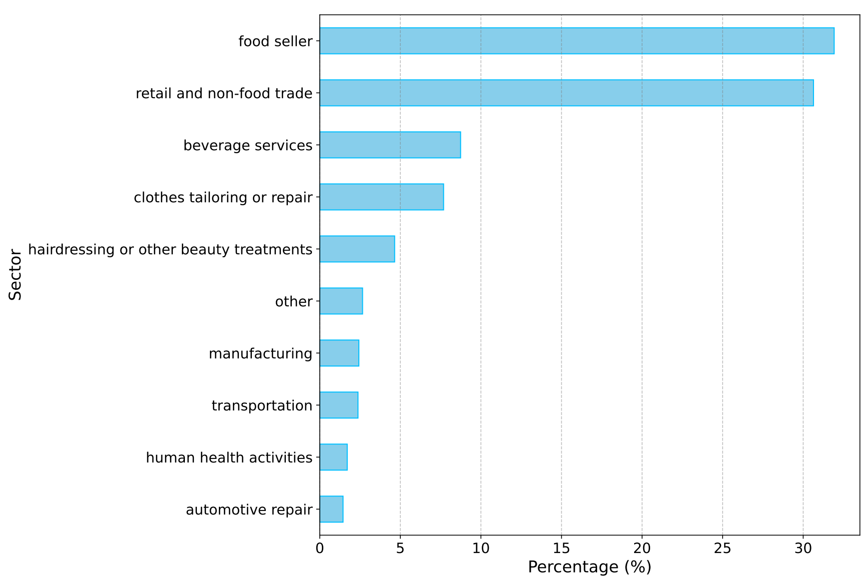
Notes: This figure plots the distribution of belief wedges among male respondents about the loan amounts requested by peers. The wedge is defined as (guess − true)/true, where *guess* is the respondent's prior belief about the average amount requested by borrowers with the same characteristics as themselves, and *true* is the model-predicted amount conditional on the respondent's characteristics. *Target women* refers to beliefs about female peers; *Target men* refers to beliefs about male peers. The sample includes all male respondents who were asked about the respective target gender. Vertical lines indicate the sample mean for each group. Wedges are winsorized at the 1st and 99th percentiles.

Figure A.5: Loan Sizes



Notes: This figure shows the proportion of loan applications from the training dataset in 6 different bins of amount size. The data consists of approximately 365,000 loan applications from May 2020 through June 2022.

Figure A.6: Top-10 Operating Sectors in the Training Data



Notes: This figure shows the proportion of loan applications from the training dataset that fall into the ten most frequent occupational sectors. The data consists of approximately 365,000 loan applications from May 2020 through June 2022. The occupations were classified into sectors according to the procedure detailed in Appendix C.

B Tables

Table B.1: Balance Table

	All (1)	Control (2)	Own gender (3)	Other gender (4)	Both (5)	$p-val$ (2)-(3)	$p-val$ (2)-(4)	$p-val$ (2)-(5)
Initial Amount	3,347	3,377	3,240	3,589	3,200	0.492	0.335	0.414
Pred. OLS Men	3,511	3,491	3,520	3,497	3,524	0.583	0.911	0.536
Pred. OLS Women	3,036	3,030	3,046	3,029	3,036	0.714	0.989	0.900
Pred. XGB Men	3,331	3,298	3,367	3,307	3,330	0.267	0.893	0.635
Pred. XGB Women	2,681	2,675	2,687	2,677	2,679	0.781	0.966	0.932
Age	38.946	39.015	38.069	39.639	39.128	0.322	0.546	0.911
Female	0.765	0.762	0.770	0.780	0.748	0.844	0.674	0.758
Week 1	0.131	0.138	0.125	0.141	0.125	0.706	0.951	0.691
Week 2	0.491	0.438	0.546	0.505	0.439	0.037	0.203	0.988
Week 3	0.378	0.423	0.328	0.355	0.436	0.056	0.174	0.800
Single	0.236	0.169	0.221	0.266	0.249	0.218	0.028	0.066
Separ./Div./Wid.	0.070	0.077	0.072	0.070	0.065	0.845	0.807	0.663
Married	0.694	0.754	0.707	0.664	0.685	0.318	0.060	0.149
Loan numb. 1	0.392	0.415	0.385	0.349	0.433	0.549	0.183	0.732
Loan numb. 2	0.234	0.262	0.239	0.251	0.199	0.610	0.812	0.148
Loan numb. 3	0.155	0.123	0.155	0.177	0.146	0.380	0.156	0.518
Loan numb. ≥ 4	0.219	0.200	0.221	0.223	0.221	0.623	0.587	0.621
Sector = Food seller	0.361	0.323	0.340	0.391	0.368	0.725	0.173	0.372
Sector = Retail and non-food trade	0.299	0.354	0.313	0.275	0.287	0.405	0.098	0.161
Sector = Rest	0.340	0.323	0.346	0.333	0.346	0.636	0.834	0.645
Region = Volta	0.082	0.085	0.081	0.101	0.062	0.887	0.595	0.397
Region = Northern	0.022	0.015	0.039	0.009	0.019	0.200	0.566	0.810
Region = Accra	0.320	0.292	0.358	0.257	0.355	0.179	0.441	0.202
Region = Northern	0.145	0.123	0.140	0.153	0.150	0.627	0.414	0.467
Region = Other South	0.432	0.485	0.382	0.480	0.414	0.044	0.931	0.173
Joint F-statistic			0.934	1.174	0.796			
F-statistic p-val			0.544	0.273	0.719			
Observations	1,113	130	335	327	321			

Notes: This table reports summary statistics and balance checks across experimental arms for the 1,113 participants in our experiment. Column (1) shows means for the full sample. Columns (2)–(5) show means by treatment assignment: control group, own-gender information, opposite-gender information, and both-gender information. For continuous variables (initial amount requested and model predictions), means are reported in Ghanaian cedis. For binary variables, means correspond to proportions. Columns (6)–(8) report p-values from two-sided t-tests of equality of means between the control group and each treatment arm. The joint F-statistic and its p-value are from a regression of a treatment indicator on all listed covariates jointly. Predictions are generated from OLS and XGBoost models trained separately by gender on historical loan applications.

Table B.2: Predictive Model Performance

	Men		Women	
	XGBoost	OLS	XGBoost	OLS
MSE	3,906,082.39	3,944,923.45	1,508,463.94	1,533,616.59
R^2	0.063	0.054	0.128	0.113

Notes: This table reports out-of-sample predictive performance of the two models used to generate objective peer information in the experiment. Both an XGBoost (Extreme Gradient Boosting) and an OLS model were trained separately for men and women on historical loan application data from the partner microcredit institution, covering 280,866 loan requests between January 2021 and June 2022. The models predict the loan amount requested by a borrower conditional on their characteristics (loan cycle, marital status, age, week, sector, and region). MSE is the mean squared error in Ghanaian cedis squared.

Table B.3: Effects of Peer Information on Borrowing Behavior: Heterogeneity by Applicant Gender

	Any change		Adjusted up		Adjusted down		Log(F/I)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Info	0.026	0.018	0.008	0.002	0.018*	0.015*	-0.000	-0.002
	(0.0477)	(0.0500)	(0.0469)	(0.0493)	(0.0093)	(0.0086)	(0.0146)	(0.0151)
Info × Female	0.052	0.057	0.060	0.062	-0.009	-0.005	0.017	0.017
	(0.0534)	(0.0557)	(0.0526)	(0.0550)	(0.0099)	(0.0095)	(0.0164)	(0.0170)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Control mean	0.046	0.046	0.046	0.046	0.000	0.000	0.014	0.014
Observations	1,113	1,113	1,113	1,113	1,113	1,113	1,113	1,113

Notes: This table summarizes the main results from providing any type of information. *Log(F/I)* is the log difference between the final and initial amount asked. All regressions include call center agent and gender fixed effects . Columns (1) to (6) include quintile of initial amount asked fixed effects. Additional controls include age, region, sector, week and loan cycle fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.4: Effects of Peer Information on Borrowing Behavior by Reference Group Gender: Heterogeneity by Applicant Gender

	Any change		Adjusted up		Adjusted down		Log(F/I)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Own gender	0.048	0.040	0.046	0.041	0.002	-0.001	0.021	0.020
	(0.0569)	(0.0594)	(0.0569)	(0.0591)	(0.0027)	(0.0040)	(0.0191)	(0.0198)
Other gender	0.058	0.051	0.003	-0.000	0.055**	0.051**	-0.020	-0.022
	(0.0586)	(0.0603)	(0.0530)	(0.0556)	(0.0269)	(0.0257)	(0.0180)	(0.0185)
Both	-0.023	-0.035	-0.025	-0.033	0.002	-0.001	-0.003	-0.005
	(0.0499)	(0.0526)	(0.0497)	(0.0521)	(0.0030)	(0.0041)	(0.0151)	(0.0157)
Own gender × Female	0.034	0.037	0.018	0.019	0.015*	0.018*	-0.005	-0.004
	(0.0643)	(0.0670)	(0.0638)	(0.0664)	(0.0084)	(0.0094)	(0.0221)	(0.0229)
Other gender × Female	0.022	0.022	0.073	0.070	-0.052*	-0.048*	0.039*	0.038*
	(0.0651)	(0.0668)	(0.0601)	(0.0624)	(0.0270)	(0.0261)	(0.0200)	(0.0203)
Both × Female	0.096*	0.109*	0.089	0.097	0.007	0.012	0.019	0.020
	(0.0579)	(0.0603)	(0.0575)	(0.0597)	(0.0071)	(0.0076)	(0.0177)	(0.0184)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
Mean dep. var.	0.106	0.106	0.096	0.096	0.010	0.010	0.026	0.026
p-val $\beta_{own} = \beta_{other}$	0.845	0.838	0.361	0.391	0.041	0.043	0.027	0.032
p-val $\beta_{own} = \beta_{both}$	0.108	0.092	0.107	0.091	0.965	0.891	0.130	0.131
p-val $\beta_{other} = \beta_{both}$	0.075	0.063	0.473	0.399	0.042	0.044	0.231	0.251
p-val $\beta_{own} + \beta_{other} = \beta_{both}$	0.079	0.093	0.285	0.295	0.041	0.048	0.865	0.898
Observations	1,113	1,113	1,113	1,113	1,113	1,113	1,113	1,113

Notes: This table summarizes the main results from the experiment. The treatment is providing information about own gender peers, opposite gender peers, or both. *Log(F/I)* is the log difference between the final and initial amount asked. All regressions include call center agent and gender fixed effects. Columns (1) to (6) include quintile of initial amount asked fixed effects. Additional controls include age and region, sector, week and loan cycle fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.5: The Effect of Amount Said on the Amount Asked: 2SLS

	Any change=1		Adjusted up=1		Adjusted down=1		Log(F/I)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
log Info said	0.684	0.657	0.180	0.204	0.503*	0.453*	-0.140	-0.116
	(0.5000)	(0.4728)	(0.4230)	(0.4082)	(0.2782)	(0.2477)	(0.1754)	(0.1601)
log Info said X Female	-0.401	-0.421	0.131	0.074	-0.532*	-0.496*	0.226	0.200
	(0.5628)	(0.5441)	(0.4920)	(0.4826)	(0.2873)	(0.2629)	(0.1953)	(0.1839)
Controls	No	Yes	No	Yes	No	Yes	No	Yes
First-stage F-stat	17.5	20.7	17.5	20.7	17.5	20.7	26.6	39.8
Observations	662	662	662	662	662	662	662	662

Notes: This table presents the results of a 2SLS regression, where the amount said is instrumented by the assignment to a high amount among individuals who received information about a single gender. *Log(F/I)* is the log difference between the final and initial amount asked. All regressions include call center agent and gender fixed effects . Columns (1) to (6) include quintile of initial amount asked fixed effects. Additional controls include age, region, sector, week and loan cycle fixed effects. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.6: Summary Table: Training Data

	Women	Men
Amount Asked	2,902	3,912
Age	40.50	39.99
Single	0.20	0.21
Separ./Div./Wid.	0.07	0.01
Married	0.71	0.77
Loan Cycle 1	0.33	0.38
Loan Cycle 2	0.23	0.21
Loan Cycle 3	0.16	0.14
Loan Cycle 4	0.11	0.10
Loan Cycle 5	0.07	0.07
Loan Cycle 6	0.04	0.04
Loan Cycle > 6	0.05	0.05
sector==agriculture or farming	0.01	0.02
sector==automotive repair	0.00	0.05
sector==beverage services	0.07	0.08
sector==clothes tailoring or repair	0.06	0.07
sector==construction	0.00	0.01
sector==creative, arts, and entertainment activities	0.01	0.01
sector==education and training	0.00	0.01
sector==fabricated metal products	0.00	0.01
sector==finance and business	0.00	0.01
sector==food seller	0.33	0.07
sector==hairdressing or other beauty treatments	0.04	0.03
sector==human health activities	0.01	0.04
sector==manufacturing	0.01	0.02
sector==mobile money	0.01	0.03
sector==other	0.14	0.23
sector==retail and non-food trade	0.30	0.19
sector==sports activities, amusement, or recreation	0.00	0.01
sector==transportation	0.00	0.08
sector==wood products	0.00	0.00
Year: 2021	0.65	0.60
Year: 2022	0.35	0.40
Observations	223745	57121

Notes: This table shows descriptive statistics (mean) for loan applicants in the training data, separating them by gender.

Table B.7: Balance table: Observed vs Non-observed in Loan Books

	All (1)	Observed (2)	Non-observed (3)	$p - val$ (2)-(3)
Initial Amount	3,347	3,370	3,231	0.399
Amount final	3,422	3,446	3,302	0.391
Age	39	39	37	0.000
Female	0.77	0.79	0.66	0.000
Week 1	0.13	0.13	0.13	0.974
Week 2	0.49	0.47	0.58	0.007
Week 3	0.38	0.40	0.29	0.006
Control	0.12	0.11	0.14	0.379
Own gender	0.30	0.30	0.30	0.947
Other gender	0.29	0.29	0.29	0.992
Both	0.29	0.29	0.27	0.585
Any Information	0.88	0.89	0.86	0.379
Single	0.24	0.23	0.27	0.216
Separ./Div./Wid.	0.07	0.07	0.09	0.327
Married	0.69	0.70	0.64	0.092
Loan numb. 1	0.39	0.33	0.73	0.000
Loan numb. 2	0.23	0.25	0.14	0.001
Loan numb. 3	0.16	0.17	0.08	0.002
Loan numb. ≥ 4	0.22	0.25	0.05	0.000
Sector = Food seller	0.36	0.37	0.29	0.036
Sector = Retail and non-food trade	0.30	0.31	0.26	0.214
Sector = Rest	0.34	0.32	0.45	0.001
Region = Volta	0.08	0.09	0.05	0.138
Region = Northern	0.02	0.02	0.04	0.092
Region = Accra	0.32	0.31	0.35	0.374
Region = Northern	0.14	0.15	0.14	0.888
Region = Other South	0.43	0.43	0.42	0.682
N	1,113	929	184	

Notes: This table shows the balance of individual characteristics between the subsample that was found in any of the three loanbooks and the subsample that was not. Means are reported in columns (1) to (3) and *p-values* are reported in column (4).

C Classifying Occupations into Sectors

C.1 Sector Classification for Experimental Sample

Table C.8 documents the guidelines given to enumerators to classify each of the loan applicants into an occupational sector during the data collection. This was necessary since the prediction model was fed with a sector. The guidelines are based on how the training dataset had previously been classified by the OpenAI's GPT-3.5 Turbo model.

Table C.8: Sector Classification Guidelines for Experimental Sample

Sector	Description	Examples/Key Words	Freq.
Agriculture or Farming	Directly extracting or farming natural resources	Poultry farmer, charcoal sprayer, poultry farm, palm oil, agro chemicals	0.8%
Automotive Repair	Vehicle maintenance and parts sale	Mechanic (sells) spare parts	1.2%
Beverage Services	Drinks sale and serving locations	Drinking spot, cold store, sells drinks, bar drinks, koko seller, restaurant	7.3%
Clothes Tailoring or Repair	Clothing creation and alterations	Seamstress, tailor, boutique, fashion designer, sells clothes, selling dresses	6.6%
Construction	Working in construction-related jobs	Carpenter, block factory, cement, cement seller	0.3%
Creative Arts and Entertainment Activities	Work in the entertainment industry or art making	Spot photographer, earrings, decoration, sells beads, jewelry spot operator	0.7%
Education and Training	Teaching and church	Pastor, teacher, school, school owner, foals teaching, church owner	0.4%
Fabricated Metal Products	Metal manipulation and sale	Welder, blacksmith, aluminum fabricator	0.3%
Finance and Business	Non-commodity trade	Lotto agent, businesswoman, businessman, salary loan, provision sales, prepaid, joint credit, vendor, startup loan	0.4%
Food Seller	Sales of any type of food	Food vendor, food joint, chop bar, food stuff, baker, sells fish, indomie, waakye, pastries, yam, fruits, porridge, eggs, meat, rice	28%
Hairdressing or Other Beauty Treatments	Hair and beauty care services	Hairdresser, cosmetics, beautician, barbering shop, salon	4%

Continued on next page

Sector	Description	Examples/Key Words	Freq.
Human Health Activities	Medicinal products and services	Pharmacy, drug store, herbal medicine	1.4%
Manufacturing	Mainly selling of foss	Sells foss	1.6%
Mobile Money	Any sales or activities associated with mobile money services	Mobile Money Vendor, Momo Vendor	1%
Other	Any occupation that is unrelated to any of the other named categories or key words in this table		15%
Printing or Reproduction of Recorded Media	Anything related to printing	Owns or operates a printing press, printing, book producer	0.2%
Retail and Non-Food Trade	Any sales that are not related to food or the exceptions listed below	Trader, provisions, sells slippers, cloth, charcoal, bags, shoes, bowls, materials, petty trader, clothing, clothes	27.5%
Sports Activities, Amusement or Recreation	Sports and recreation	Drinking sport, sport operator, lottery ticket, game center	0.3%
Transportation	Any carriage of persons	Taxi driver, car owner, transport	2%
Wood Products	Selling wood	Sells wood or timber	0.1%

C.1.1 Remarks

- Sales of food go into “food seller”, sales of anything other than food go into “retail and non-food trade”.
- Exceptions:
  1. Selling of drinks goes into beverage services.
  2. Selling of clothes or clothing can fit into both “clothes tailoring and repair” and “retail and non-food trade”. In this special case the call center employee should inquire more, otherwise if in doubt put it into “retail and non-food trade”. NOTE: selling of “cloth” ALWAYS goes into “retail and non-food trade”.
  3. Mobile money sales go under “Mobile Money”.
  4. Sells foss goes under “manufacturing”.

- Note: Provisions or Provision shop goes under retail.

D Experimental Procedures

D.1 Phone script

1. Ask about occupation.
2. Ask about employment situation in order to determine their loan type.
3. Talk about home/business accessibility.
4. Walk them through terms and conditions.
5. Take personal details including age, marital status, region and occupation.
6. Record initial ask amount.
7. Talk them through repayment conditions (including interest rate).
8. Ask them how much they think other men and women with the same characteristics would usually ask for:

> **Control [randomized order]:** *To help us improve our services to you, we would like to ask you an additional question: Please tell me, if you have to guess, how much money do you think women (men) who are similar to you usually request in a loan at the microcredit institution? By similar, we mean the same line of work, age, region, marital status, and also applying for the loan for the first time like you. What about men (women) who are similar to you?*
>
> **T1 (female info only):** *To help us improve our services to you, we would like to ask you an additional question: Please tell me, if you have to guess, how much money do you think women who are similar to you usually request in a loan at the microcredit institution? By similar, we mean the same line of work, age, region, marital status, and also applying for the loan for the first time like you.*
>
> **T2 (male info only):** *To help us improve our services to you, we would like to ask you an additional question: Please tell me, if you have to guess, how much money do you think men who are similar to you usually request in a loan at the microcredit institution? By similar, we mean the same line of work, age, region, marital status, and also applying for the loan for the first time like you.*

T3 (both) [randomized order]: *To help us improve our services to you, we would like to ask you an additional question: Please tell me, if you have to guess, how much money do you think women (men) who are similar to you usually request in a loan at the microcredit institution? By similar, we mean the same line of work, age, region, marital status, and also applying for the loan for the first time like you? What about men (women) who are similar to you.*

9. Inform Them About Predicted Amounts:

   Control: [nothing]

   T1 (female info only): *Women who are similar to you usually request approximately X cedis. This is just an extra piece of information for you. You are free to request any amount you think is appropriate for you.*

   T2 (male info only): *Men who are similar to you usually request approximately X cedis. This is just an extra piece of information for you. You are free to request any amount you think is appropriate for you.*

   T3 (both) [randomized order]: *Men who are similar to you usually request approximately X cedis. Women who are similar to you usually request approximately Z cedis. This is just an extra piece of information for you. You are free to request any amount you think is appropriate for you.*

10. Get New Ask Amount:
    Now, just for confirmation: do you confirm the Z amount you already asked, or do you wish to revise the required amount?